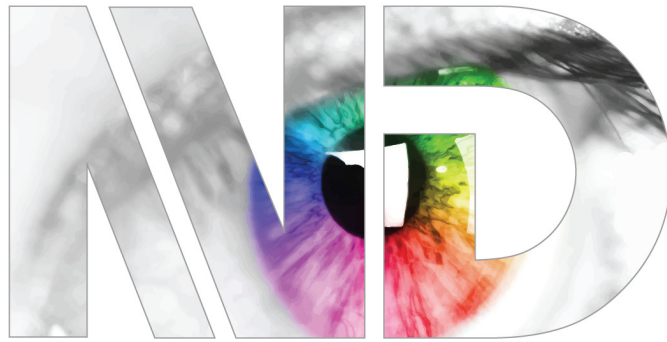




New Vision Display

PRODUCT SPECIFICATION

1.44" 128RGB x 128 TFT

RFQ NUMBER: EM-49906-F

MODEL NUMBER: 1M.DYSO-9A88DPOP014H-01

Rev: 05

CUSTOMER P/N: 725536-01/06

New Vision Display	PREPARED BY	CHECKED BY	APPROVED BY
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DATE	2023-12-15	2023-12-15	2023/12/15

CUSTOMER APPROVAL	SIGNATURE	DATE

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Revision History

[illegible]

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1. General Description

This display module consists of a transmissive 1.44 inch 128BGR x 128, TFT a-Si Active Matrix Color LCD that is electronically and mechanically integrated. The TFT display is capable of displaying 262K colors. Each pixel is divided into Red, Green and Blue sub-pixels or dots which are arranged in vertical stripes.

The display drivers incorporated is the ST7735P3, The ST7735P3 is a single-chip controller/driver for 262K-color, graphic type TFT-LCD. It consists of 396 source line and 162 gate line driving circuits. This chip is capable of connecting directly to an external microprocessor, and accepts Serial Peripheral Interface (SPI), 8-bit/9-bit/16-bit/18-bit parallel interface. Display data can be stored in the on-chip display data RAM of 132 x 162 x 18 bits. It can perform display data RAM read/write operation with no external operation clock to minimize power consumption.

2. Module Parameter

Features	Details	Unit
Display Size(Diagonal)	1.44"	Inch
LCD type	α-Si TFT	-
Display Mode	IPS / Transmissive / Normally black	-
Resolution	128 BGR(H) x 128	-
View Direction	All Direction	-
Grayscale Reversion	N/A	
Module Outline	40.0(H) x 33.94(V) x 7.0 (Note 1)	mm
Top Surface Polarizer	Anti-Glare	
Active Area	25.4976(W) x 26.496(H)	mm
Pixel Size	0.1992(H) x 0.207(V)	mm
Interface	4-Wire Serial	-
Pixel arrangement	BGR vertical stripe	
Display Color	262K(18bit)	
Driver IC	ST7735P3	-
Back-light	3 White LED*1link	-
Power Consumption	53.73 (typ.) / 61.488 (max.)	mW (Note2)
Weight	5.8 (typ.)	g

Note 1: Excluding hooks, posts, FPC tail, black Tape etc.

Note2: it includes the power consumption of TFT driver and backlight light, for details, please refer to Section 4&5, Electrical Characteristics & Backlight Characteristics sections.

3. Absolute Maximum Ratings

GND=0V, Ta=25±2°C

Item	Symbol	Min.	Max.	Unit
Display Driver Power Supply	VDD	-0.3	4.6	V
I/O Power Supply	VDDIO	-0.3	4.6	V
Storage temperature	Tstg	-30	+80	°C
Operating temperature	Topa	-20	+70	°C
Storage humidity	Hstg	10	Note 1	%RH
Operating humidity	Hopa	10	Note 1	%RH

Note: 1. Note 1: 90% max, if Ta is below 50 deg C; 60% max, if Ta is over 50 deg C.

2. If the absolute maximum rating of even is one of the above parameters is exceeded even momentarily, the quality of the product may be degraded. Absolute maximum ratings, therefore, specify the values exceeding which the product may be physically damaged. Be sure to use the product within the range of the absolute maximum ratings.

4. Electrical Characteristics

Item	Symbol	Min.	Typ.	Max.	Unit
Display Driver Power Supply	VDD	3.1	3.3	3.465	V
I/O Power Supply(Note1)	VDDIO	3.1	3.3	3.465	V
Logic Low input voltage	V _{IL}	GND	-	0.3* VDDIO	V
Logic High input voltage	V _{IH}	0.7*VDDIO	-	VDDIO	V
Logic Low output voltage	V _{OL}	GND	-	0.2*VDDIO	V
Logic High output voltage	V _{OH}	0.8*VDDIO	-	VDDIO	V
Current Consumption (at black display pattern)	I _{DD} + I _{DDIO}	-	2.1	3.2	mA
Frame Frequency	f _{FR}	-	60	-	Hz

Note: 1.VDDIO≤VDD

5. Backlight Characteristics

Item	Symbol	Condition	Min	Typ	Max	Unit
Forward Voltage	V _f	T _a =25 °C, I _f =18mA	-	2.6	2.8	V
Forward Current	I _f	T _a =25 °C	-	18	-	mA
Reverse Voltage	V _R	-	-	-	5	V
Reverse current	I _R	-	--	-	50	uA
Power dissipation	P _d		-	46.8	50.4	mW
LED presumption of life time(Note)			80000	-	-	Hours
Drive method	Constant current					
LED Configuration	1 White LED* 3link					

Note: The LED presumption life time is defined as the module surface luminance decrease to 50% of the original luminance when the ambient temperature is 25 °C and operating current is 18mA, the LED life time would decrease if the operating current is larger than 18mA. And it's presumed by LED supplier.

6. Optical Characteristics

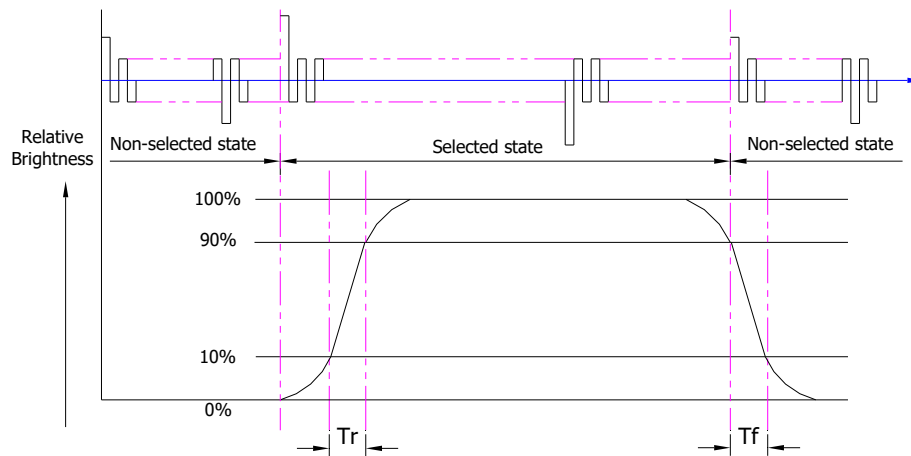
6.1. Optical Characteristics

T_a=25°C, VDD=3.3V, ILED=18mA

Backlight On (Transmissive Mode)	Item		Symbol	Condition	Specification			Unit
					Min.	Typ.	Max.	
	Contrast ratio		CR	θ= 0° Normal viewing angle	800	1000	-	cd/m ²
	Surface Luminance		L _V		300	350	680	
	Response time		T _F +T _R		-	35	40	
	Color Chromaticity	Red	X _R		0.56	0.59	0.62	-
			Y _R		0.31	0.34	0.37	
		Green	X _G		0.33	0.36	0.39	
			Y _G		0.54	0.57	0.60	
		Blue	X _B		0.12	0.15	0.18	
			Y _B		0.08	0.11	0.14	
		White	X _W		0.28	0.31	0.34	
			Y _W		0.29	0.32	0.35	
	Viewing Angle	Horizontal	θ _L	CR>10	75	85	-	-
θ _R			75		85	-		
Vertical		θ _U	75		85	-		
		θ _D	75		85	-		
Uniformity					75	-	-	%
NTSC Ratio(Gamut)					45	55		%

6.2. Definition of Response Time

6.2.1. Normally Black Type (Negative)

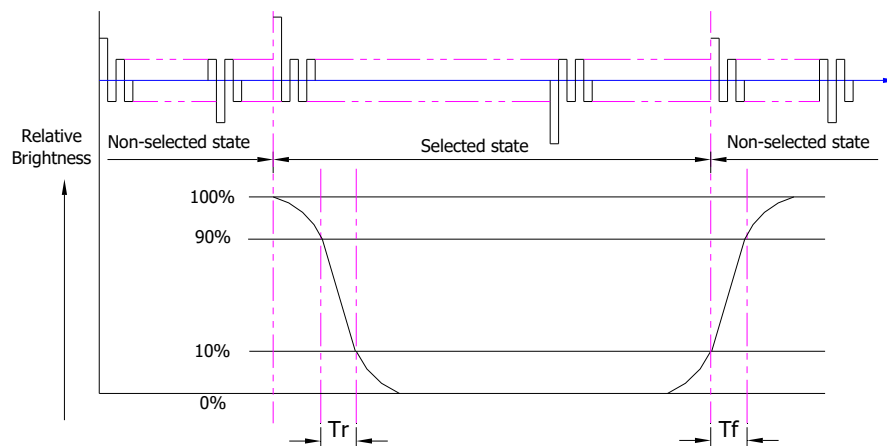


Tr is the time it takes to change from non-selected state with relative luminance 10% to selected state with relative luminance 90%;

Tf is the time it takes to change from selected state with relative luminance 90% to non-selected state with relative luminance 10%.

Note: Measuring machine: BM-7

6.2.2. Normally White Type (Positive)



Tr is the time it takes to change from non-selected state with relative luminance 90% to selected state with relative luminance 10%;

Tf is the time it takes to change from selected state with relative luminance 10% to non-selected state with relative luminance 90%;

6.3. Definition of Contrast Ratio

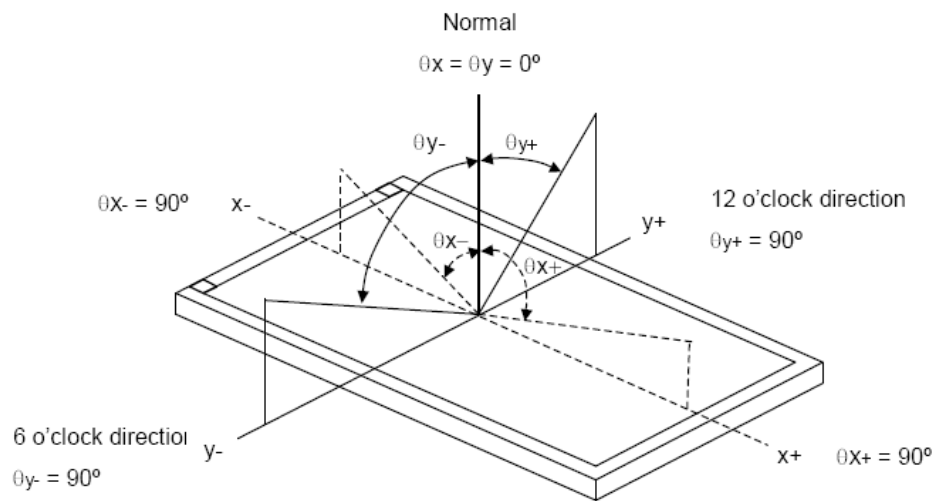
Contrast is measured perpendicular to display surface in reflective and transmissive mode.

The measurement condition is:

Measuring Equipment	BM-7 or Equivalent
Measuring Point Diameter	3mm//1mm
Measuring Point Location	Active Area centre point
Test pattern	A: All Pixels white
	B: All Pixel black
Contrast setting	Maximum

Definitions: CR (Contrast) = Luminance of White Pixel / Luminance of Black Pixel

6.4. Definition of Viewing Angles



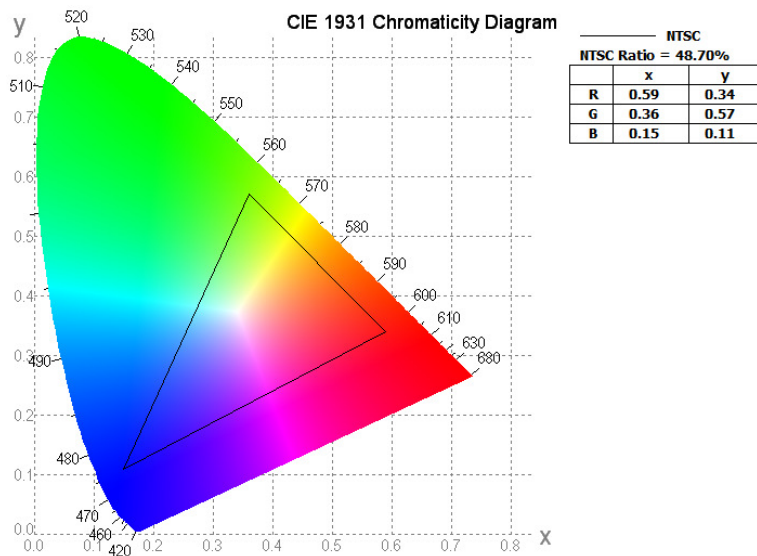
Measuring machine: LCD-5100 or EQUI

6.5. Definition of Color Appearance

R, G, B and W are defined by (x, y) on the CIE chromaticity diagram

NTSC=area of RGB triangle/area of NTSC triangleX100%

Measuring picture: Red, Green, Blue and White (Measuring machine: BM-7)

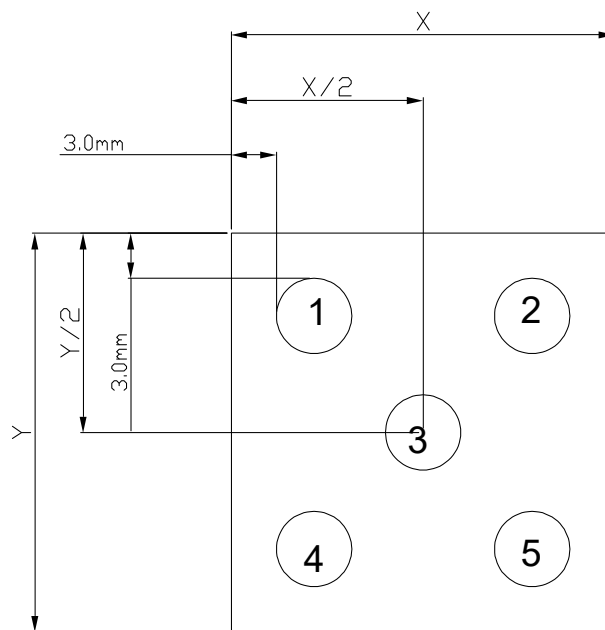


Definition of Surface Luminance, Uniformity and Transmittance

Using the transmissive mode measurement approach, measure the white screen luminance of the display panel and backlight.

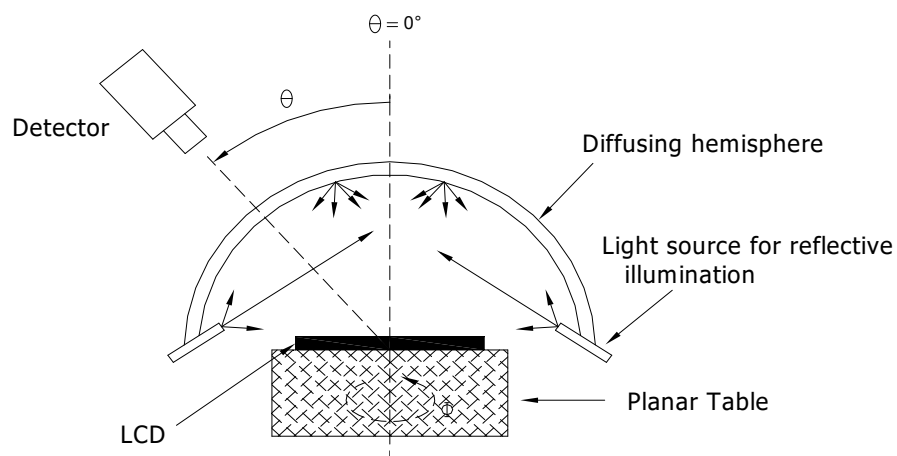
- 5.6.1 Surface Luminance: $L_V = \text{average } (L_{P1}:L_{P5})$
- 5.6.2 Uniformity = $\text{Minimal } (L_{P1}:L_{P5}) / \text{Maximal } (L_{P1}:L_{P5}) * 100\%$
- 5.6.3 Transmittance = $L_V \text{ on LCD} / L_V \text{ on Backlight} * 100\%$

Note: Measuring machine: BM-7 OR EQUIVALENT

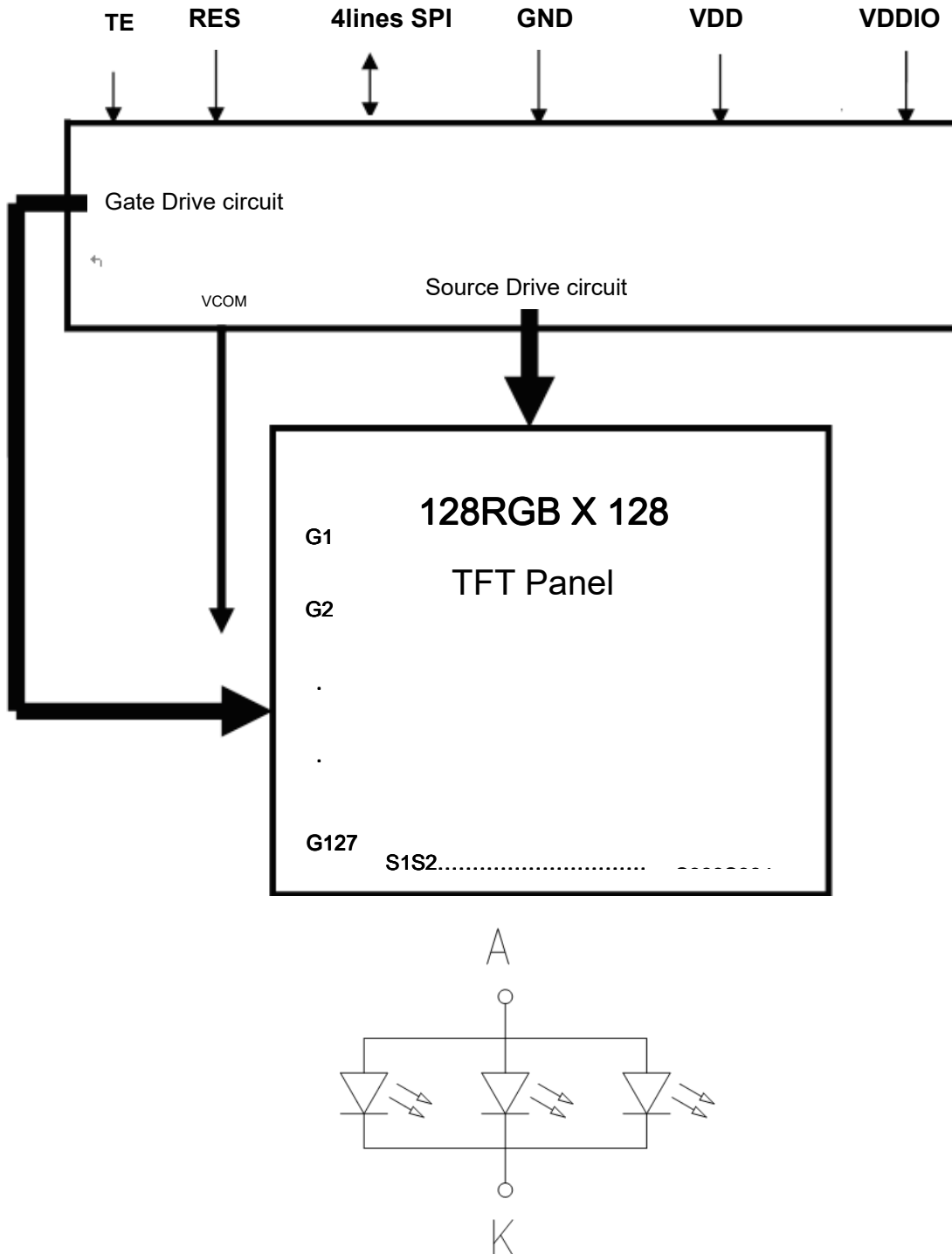


6.6. Definition of Reflectivity

To measure the reflectivity, the detector should be aligned to the normal direction of the LCD surface corresponding azimuthally angle $\theta=0^\circ$



7. Block Diagram and Power Supply



SCHEMATIC FOR LED CIRCUIT
CONSTANT CURRENT

8. Interface Pins Definition

No.	Symbol	I/O	Function
1	A	P	LED Anode
2	K	P	LED Cathode
3	TE	O	Tearing status
4	VDD	P	3.3V Display Driver Power Supply
5	VDDIO	P	3.3V I/O Power Supply
6	GND	P	Ground
7	SCL	I	Serial Interface Clock Signal
8	RESETX	I	Active Low Reset signal
9	CSX	I	Active Low Chip Select
10	SDA	I/O	Serial Interface Data Signal
11	D/C	I	Data/Command Signal
12	GND	P	Ground

9. Display AC Characteristics

9.1. Power Sequence

VDDIO and VDD can be applied in any order

VDD and VDDIO can be powered down in any order

During power off, if LCD is in the Sleep Out mode, VDD and VDDI must be powered down minimum 120msec

after RESX has been released.

During power off, if LCD is in the Sleep In mode, VDDI or VDD can be powered down minimum 0msec after

RESX has been released.

CSX can be applied at any timing or can be permanently grounded. RESX has priority over CSX.

Note 1: There will be no damage to the display module if the power sequences are not met.

Note 2: There will be no abnormal visible effects on the display panel during the Power On/Off Sequences.

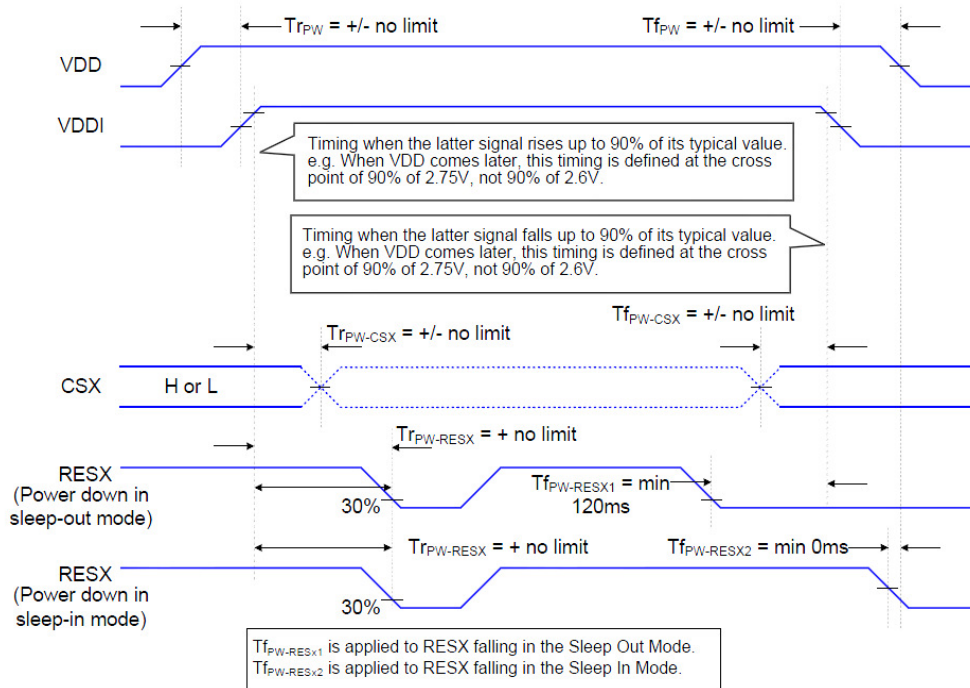
Note 3: There will be no abnormal visible effects on the display between end of Power On Sequence and before receiving Sleep Out

command. Also between receiving Sleep In command and Power Off Sequence.

Note 4: If RESX line is not held stable by host during Power On Sequence as defined in the sequence below, then it will be necessary to

apply a Hardware Reset (RESX) after Host Power On Sequence is complete to ensure correct operation. Otherwise function is not guaranteed.

The power on/off sequence is illustrated below



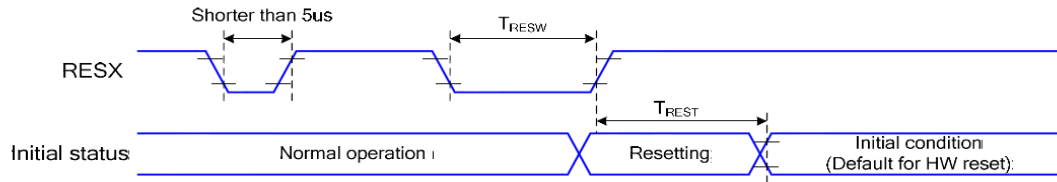
Note1: Unless otherwise specified, timings herein show cross point at 50% of signal power level.

9.1.1. Uncontrolled Power Off Sequence

The uncontrolled power-off means a situation which removed a battery without the controlled power offsequence. It will neither damage the module or the host interface.

If uncontrolled power-off happened, the display will go blank and there will not any visible effect on the display(blank display) and remains blank until "Power On Sequence" powers it up.

9.2. Reset Timing



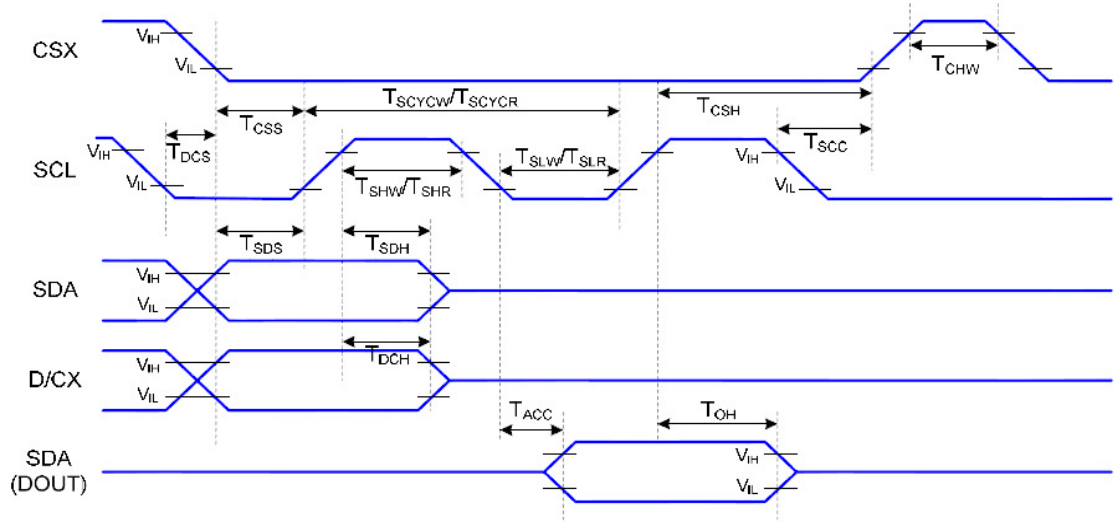
Related Pins	Symbol	Parameter	MIN	MAX	Unit
RESX	tRESW	Reset Pulse Duration	10	-	us
	tREST	Reset Cancel	-	5	ms
				120	ms

Table 14 Reset Timing

Notes:

1. The reset cancel includes also required time for loading ID bytes, VCOM setting and other settings from NVM (or similar device) to registers. This loading is done every time when there is HW reset cancel time (tRT) within 5 ms after a rising edge of RESX.
2. Spike due to an electrostatic discharge on RESX line does not cause irregular system reset according to the table below:

9.3. Serial Interface Characteristics (4-line Serial)



T_a=25 °C, V_{DDI}=1.65~3.7V, V_{DD}=2.5~4.8V

Signal	Symbol	Parameter	MIN	MAX	Unit	Description
CSX	TCSS	Chip Select Setup Time (Write)	45		ns	
	TCSH	Chip Select Hold Time (Write)	45		ns	
	TCSS	Chip Select Setup Time (Read)	60		ns	
	TSCC	Chip Select Hold Time (Read)	65		ns	
	TCHW	Chip Select "H" Pulse Width	40		ns	
SCL	TSCYCW	Serial Clock Cycle (Write)	66		ns	-Write Command & Data Ram
	TSHW	SCL "H" Pulse Width (Write)	15		ns	
	TSLW	SCL "L" Pulse Width (Write)	15		ns	
	TSCYCR	Serial Clock Cycle (Read)	150		ns	-Read Command & Data Ram
	TSHR	SCL "H" Pulse Width (Read)	60		ns	
	TSLR	SCL "L" Pulse Width (Read)	60		ns	
D/CX	TDCS	D/CX Setup Time	10		ns	
	TDCH	D/CX Hold Time	10		ns	
SDA (DIN) (DOUT)	TSDS	Data Setup Time	10		ns	For Maximum CL=30pF For Minimum CL=8pF
	TSDH	Data Hold Time	10		ns	
	TACC	Access Time	10	50	ns	
	TOH	Output Disable Time	15	50	ns	

10. Recommended Setting and Initialization Flow

//9A88 init code Rev01

//first issue

HW_Reset();

Delay(120); //Delay 120ms

Write(Command , 0x11); //Sleep out

Delay(120); //Delay 120ms

Write(Command , 0xB1);

Write(Parameter , 0x05);

Write(Parameter , 0x4C);

Write(Parameter , 0x4C);

Write(Command , 0xB2);

Write(Parameter , 0x05);

Write(Parameter , 0x4C);

Write(Parameter , 0x4C);

Write(Command , 0xB3);

Write(Parameter , 0x05);

Write(Parameter , 0x4C);

Write(Parameter , 0x4C);

Write(Parameter , 0x05);

Write(Parameter , 0x4C);

Write(Parameter , 0x4C);

Write(Command , 0xB4); //Dot inversion

Write(Parameter , 0x03);

Write(Command , 0xC0);

Write(Parameter , 0xAA);

Write(Parameter , 0x0A);

Write(Parameter , 0x84);

Write(Command , 0xC1);

Write(Parameter , 0xD5);

Write(Command , 0xC2);

Write(Parameter , 0x0D);

```
Write(Parameter , 0x00);

Write(Command , 0xC3);
Write(Parameter , 0x8D);
Write(Parameter , 0x2A);

Write(Command , 0xC4);
Write(Parameter , 0x8D);
Write(Parameter , 0xEE);

Write(Command , 0xC5);
Write(Parameter , 0x05);

Write(Command , 0x35);
Write(Parameter , 0x00);

Write(Command , 0x36);
Write(Parameter , 0x1C); //inverse H,V refresh order & RGB order for inverse view

Write(Command , 0xE0);
Write(Parameter , 0x09);
Write(Parameter , 0x16);
Write(Parameter , 0x03);
Write(Parameter , 0x05);
Write(Parameter , 0x0D);
Write(Parameter , 0x04);
Write(Parameter , 0x00);
Write(Parameter , 0x04);
Write(Parameter , 0x08);
Write(Parameter , 0x0E);
Write(Parameter , 0x1A);
Write(Parameter , 0x39);
Write(Parameter , 0x00);
Write(Parameter , 0x05);
Write(Parameter , 0x04);
Write(Parameter , 0x3F);

Write(Command , 0xE1);
Write(Parameter , 0x0B);
Write(Parameter , 0x15);
Write(Parameter , 0x04);
Write(Parameter , 0x07);
Write(Parameter , 0x0B);
Write(Parameter , 0x02);
```

```
Write(Parameter , 0x00);
Write(Parameter , 0x01);
Write(Parameter , 0x05);
Write(Parameter , 0x0C);
Write(Parameter , 0x1B);
Write(Parameter , 0x37);
Write(Parameter , 0x00);
Write(Parameter , 0x07);
Write(Parameter , 0x01);
Write(Parameter , 0x3F);

Write(Command , 0x3A);    //262k mode
Write(Parameter , 0x06);

Write(Command , 0x29);    //Display on

////////define_window////////
Write(Command , 0x2A);
Write(Parameter , 0x00);
Write(Parameter , 0x00);
Write(Parameter , 0x00);
Write(Parameter , 0x7F);

Write(Command , 0x2B);
Write(Parameter , 0x00);
Write(Parameter , 0x20);
Write(Parameter , 0x00);
Write(Parameter , 0x9F);

Write(Command , 0x2C);
////////define_window////////
```

11. Quality Assurance

11.1. Purpose:

The purpose of this specification is to establish the cosmetic standards for inspection and measurement of a NVD TFT Sub-Assembly.

11.2. Scope:

This specification applies to all TFT Sub-Assemblies built by NVD and should be used as the inspection guideline for quality control. The individual drawing specification will have priority if this document conflicts with the drawing.

11.3. Standard for Quality Test:

New Vision Display performs the following tests to ensure the quality of product before shipment.

11.3.1. Sampling Plan:

11.3.1.1. ANSI / ASQC Z1.4-2008. General Inspection Level II.

11.3.1.2. Single sampling, normal inspection.

11.3.2. Sampling Criteria:

11.3.2.1. Visual inspection: AQL 1.0

11.3.2.2. Electrical functional: AQL 0.65

11.3.3. Reliability Test:

11.3.3.1. Detailed requirement refer to Reliability Test Specification.

11.4. Nonconforming Analysis & Disposition

11.4.1. Nonconforming analysis:

11.4.1.1. Customer should provide overall information of non-conforming sample for their complaints.

11.4.1.2. After receipt of detailed information from customer, the analysis of nonconforming parts usually should be finished in one week.

11.4.1.3. If New Vision Display can not finish the analysis on time, customer will be notified with the progress status.

11.4.2. Disposition of nonconforming:

Non-conforming product over ppm level, New Vision Display will offer corrective actions, not over PPM, New Vision Display can offer FA if customer need. And the failures are confirmed to be New Vision Display responsibility and within the shelf life of 1 year, they will be replaced.

11.5. Agreement Items

New Vision Display and customer shall negotiate if the following situation occurs:

11.5.1. There is any discrepancy in standard of quality assurance.

11.5.2. Additional requirement to be added in product specification.

11.5.3. Any other special problem.

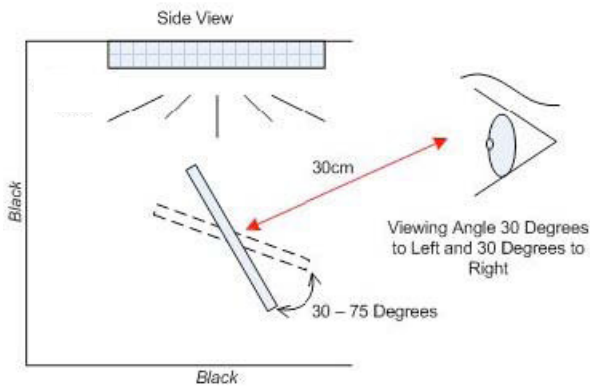
11.6. Standard Viewing Conditions

For the purpose of cosmetic inspection, all NVD touch panels are to be clean, dry

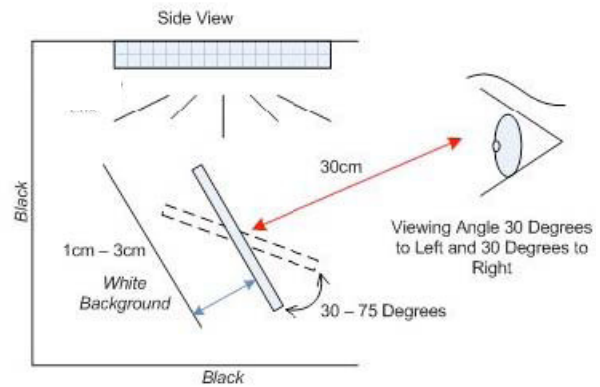
and viewed under the following conditions:

- 11.6.1. Lighting: Daylight or cool white fluorescent lighting approximately from 70 to 140 foot-candles (approximately equal to 750 to 1500 Lux). Lighting should be diffused so that shadow is not a factor.
- 11.6.2. Distance: 12 inches (approximately equal to 30~40 cm) from the eyes.
- 11.6.3. Inspection Time: 10 seconds
- 11.6.4. Inspector qualification: Inspection must be made by an individual with 20/20 or corrected to 20/20 vision with color discrimination capability.
- 11.6.5. Parts shall not be manipulated to reflect a single light source in order to accentuate surface flaws. Parts shall be viewed without directly reflecting a light source.
- 11.6.6. Please refer below photos for the inspection environment.

Black Booth or Black Background



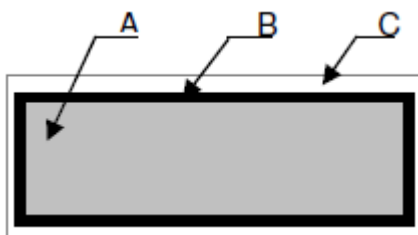
White Background



11.7. Cosmetic Specifications and Acceptance Criteria Guidelines

- 11.7.1. Defects must be found at Incoming Quality Control (IQC), prior to non-NVD TFT display bonding, or before final manufacturing assembly.
- 11.7.2. Bubbles or contamination outside the viewing area are acceptable. These anomalies do not impact functionality, performance or long term reliability.
- 11.7.3. Bubbles between TFT and protective film are acceptable. These anomalies do not impact functionality, performance or long term reliability.
- 11.7.4. If a surface blemish or defect can be wiped off easily, removed by cleaning or blown away using a compressed air gun, the TFT is acceptable.
- 11.7.5. Blemishes or defects on the TFT back side that are not visible from the front are acceptable. These anomalies do not impact performance, functionality or long term reliability.
- 11.7.6. Glass chips that do not impact functionality, performance or long term reliability and only observed from the back side are acceptable.
- 11.7.7. Glass cracks or fractures are not acceptable. This is a defect.
- 11.7.8. The FPC/PCBA refer to the IPC-A-610 (class 2) or IPC-6013.

11.8. Product surface area definition



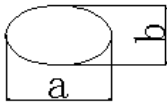
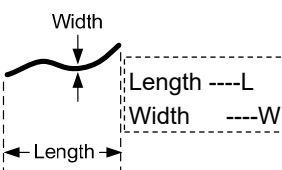
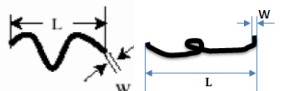
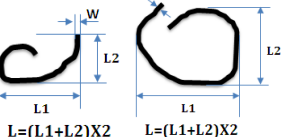
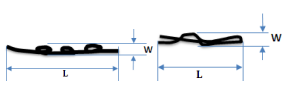
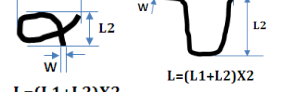
A: Display Area

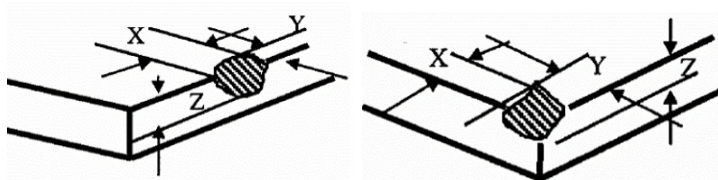
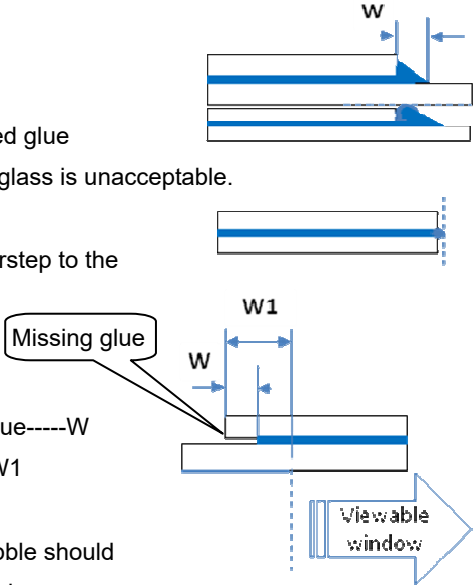
B: Viewing Area

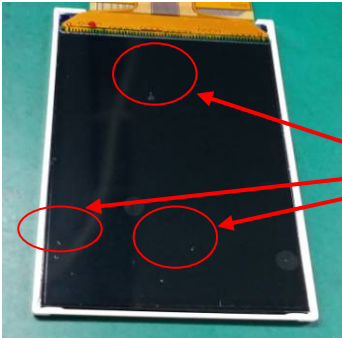
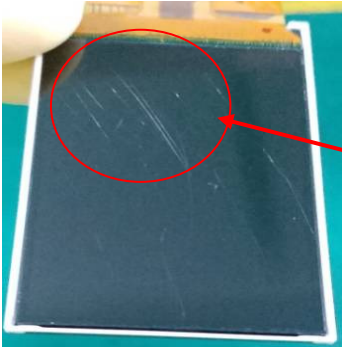
C: Out of viewing area

11.9. Inspection items and acceptance criteria

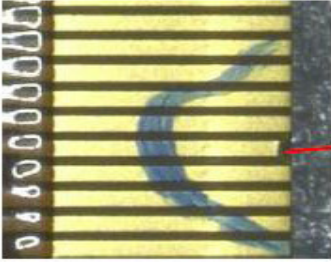
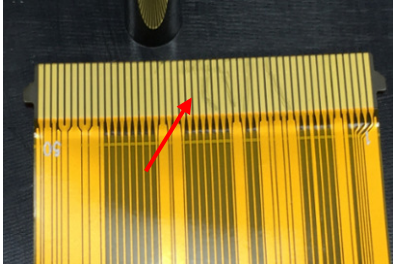
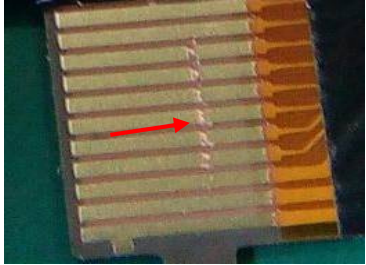
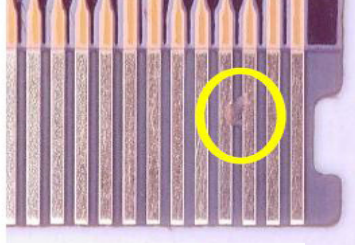
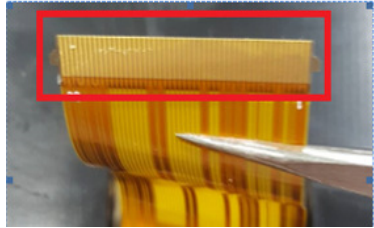
11.9.1. Visual defect

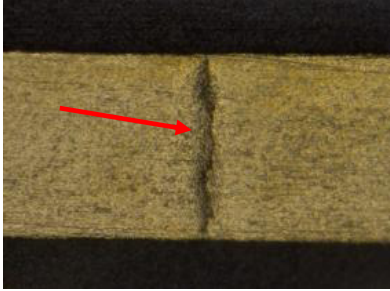
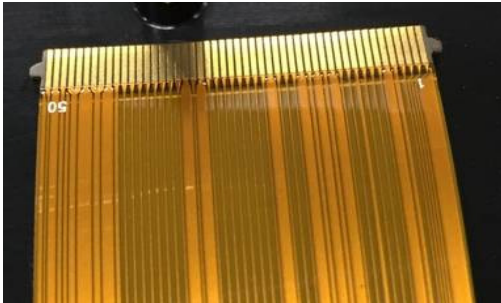
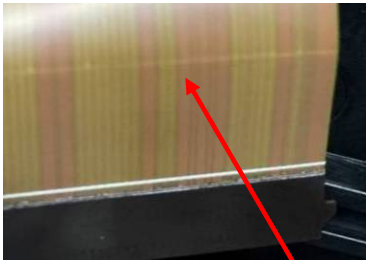
Item #	Inspection Item	Acceptance Criteria				
01	Circular defect (Include contamination, black or white dots/spots, pin hole, bubble etc.)  $\Phi=(a+b) / 2$	9.1.1 A area circular defect:				
		Diameter Φ (mm)		Accept. qty. within A area	Minimum distance between blemishes	
		Dots or Spots	$\Phi \leq 0.10$	Ignored	/	
			$0.1 < \Phi \leq 0.15$	2	3 mm	
$0.15 < \Phi \leq 0.20$	1		3 mm			
		$\Phi > 0.20$	0	/		
		9.1.2 B/C area circular defect:				
		Diameter Φ (mm)		Accept. qty. within B area	Minimum distance between blemishes	
		Dots or Spots	$\Phi \leq 0.1$	Ignored	/	
			$0.1 < \Phi \leq 0.2$	2	3 mm	
$\Phi > 0.2$	0		/			
02	Linear fibers, scratches, etc.      $L=(L1+L2) \times 2$	Linear defects:				
		Length (mm)	Width (mm)	Accept. Quantity within A area	Accept. Quantity within B area	Minimum distance between blemishes
		Ignored	$W \leq 0.03$ mm	Ignored		/
		$L \leq 5$ mm	$0.03 < W \leq 0.05$	3	2	3 mm
		$L \leq 5$ mm	$0.05 < W \leq 0.10$	2	1	3 mm
		$L > 5$ mm	$W > 0.10$	0	0	/
		Remarks:				
		a) Bubbles or contamination outside viewing area is acceptable.				
		b) It is acceptable if any above defects can be wiped off, or blown away by using a compressed air gun.				
		c) It is acceptable if any above blemish or defects only can be seen from the rear view (the blemish or defects can't be seen from the front view).				

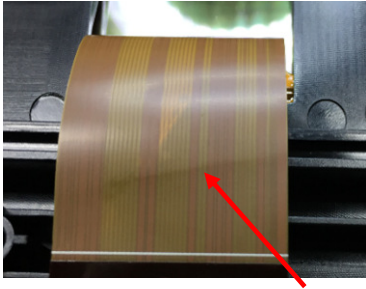
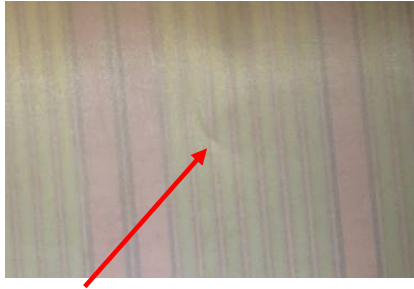
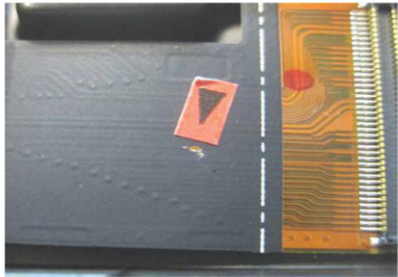
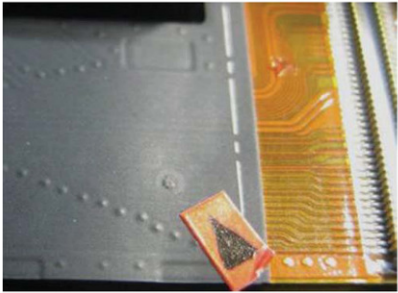
Item #	Inspection Item	Acceptance Criteria																				
03	Polarizer bubble	Diameter Φ (mm)	Accept. qty. within A area	Minimum distance between blemishes																		
		$\varphi \leq 0.2$	Ignored	/																		
		$0.2 < \varphi \leq 0.3$	3	3mm																		
		$0.3 < \varphi$	0	/																		
04	Dent	Diameter Φ (mm)	Accept. qty. within A area	Minimum distance between blemishes																		
		$\varphi \leq 0.2$	Ignored	/																		
		$0.2 < \varphi \leq 0.3$	3	3mm																		
		$0.3 < \varphi$	0	/																		
05	Glass chips on the edge	a) Chipping on the TFT display :																				
																						
		<table><tr><th colspan="3">Chip on edge</th><th colspan="3">Chip on corner</th></tr><tr><th>X</th><th>Y</th><th>Z</th><th>X</th><th>Y</th><th>Z</th></tr><tr><td>Ok $\leq$ 4mm</td><td>Ok $\leq$ 2mm</td><td>Ok $\leq$ t</td><td>Ok $\leq$ 4mm</td><td>Ok $\leq$ 3mm</td><td>Ok $\leq$ t</td></tr></table>			Chip on edge			Chip on corner			X	Y	Z	X	Y	Z	Ok $\leq$ 4mm	Ok $\leq$ 2mm	Ok $\leq$ t	Ok $\leq$ 4mm	Ok $\leq$ 3mm	Ok $\leq$ t
		Chip on edge			Chip on corner																	
X	Y	Z	X	Y	Z																	
Ok $\leq$ 4mm	Ok $\leq$ 2mm	Ok $\leq$ t	Ok $\leq$ 4mm	Ok $\leq$ 3mm	Ok $\leq$ t																	
Remarks: a. Edge chipping cannot be close to the ITO trace/PAD. Chipping if present will not impact function, performance or reliability.																						
06	Glue Defect	a) Exceed glue:																				
		1) Width of exceed glue----W W $\leq$ 2 mm																				
		2) The thickness of the exceed glue higher than smaller glass is unacceptable.																				
		3) The exceed glue can't overstep to the edge of the glass.																				
06	Glue Defect	b) Missing glue:																				
		Width of missing glue----W Width of frame----W1 W $\leq$ 1/2 W1																				
		Remarks: The large air bubble should be looked as missing glue and can't extend to viewable window.																				
																						

07	Protective Film	a) Protective film misalignment $\leq 2\text{mm}$ from edge should be acceptable. b) Protective film bubbles are acceptable.  Protective film bubbles ---Accept c) Protective film contamination: No impact the part surface under the protective film should be acceptable.  Protective film contamination ---Accept d) Protective film surface scratch: No hurt the part under the protective film should be acceptable.  Protective film surface scratch ---Accept e) Protective film broken to expose the part under the protective film, not allowed.
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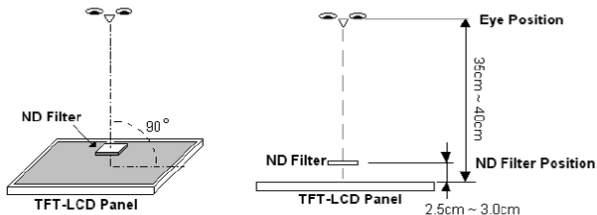
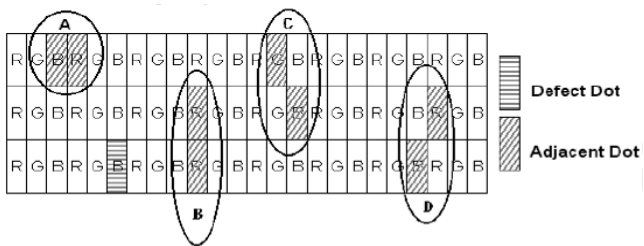
8). FPC Cosmetic Inspection Criteria:

1)	The tip of the terminal missing	  Terminal missing and perk ---Not allowed <table border="1"> <thead> <tr> <th>Item</th><th>Accept Qty</th><th>Remark</th></tr> </thead> <tbody> <tr> <td>Chip remain on the PIN</td><td>Not Allowed</td><td>Visual</td></tr> <tr> <td>Debris on the remaining PIN</td><td>Not Allowed</td><td>Visual</td></tr> <tr> <td>Terminal missing and perk</td><td>Not Allowed</td><td>Visual</td></tr> </tbody> </table>	Item	Accept Qty	Remark	Chip remain on the PIN	Not Allowed	Visual	Debris on the remaining PIN	Not Allowed	Visual	Terminal missing and perk	Not Allowed	Visual
Item	Accept Qty	Remark												
Chip remain on the PIN	Not Allowed	Visual												
Debris on the remaining PIN	Not Allowed	Visual												
Terminal missing and perk	Not Allowed	Visual												
2)	Scratch on golden finger	  Not copper exposed---Accept Copper exposed---Not allowed <table border="1"> <thead> <tr> <th>Item</th><th>Accept Qty</th><th>Remark</th></tr> </thead> <tbody> <tr> <td>Scratches on the golden fingers but no copper layer exposed</td><td>Ignore</td><td>Check by 10 time Microscope</td></tr> <tr> <td>Damage the golden layer to make the copper layer exposed</td><td>Not allowed</td><td>Check by 10 time Microscope</td></tr> </tbody> </table>	Item	Accept Qty	Remark	Scratches on the golden fingers but no copper layer exposed	Ignore	Check by 10 time Microscope	Damage the golden layer to make the copper layer exposed	Not allowed	Check by 10 time Microscope			
Item	Accept Qty	Remark												
Scratches on the golden fingers but no copper layer exposed	Ignore	Check by 10 time Microscope												
Damage the golden layer to make the copper layer exposed	Not allowed	Check by 10 time Microscope												
3)	Contamination or glue remained on the golden finger	  Nonconductive particles remained---OK Glue remained on golden fingers--- Not Allowed <table border="1"> <thead> <tr> <th>Item</th><th>Accept Qty</th><th>Remark</th></tr> </thead> <tbody> <tr> <td>Conductive contamination exist in PINs</td><td>Not Allowed</td><td>Visual</td></tr> <tr> <td>No Moving nonconductive foreigner contamination between two pins</td><td>Ignore</td><td>Visual</td></tr> <tr> <td>Glue remained on the golden finger</td><td>Not Allowed</td><td>Visual</td></tr> </tbody> </table>	Item	Accept Qty	Remark	Conductive contamination exist in PINs	Not Allowed	Visual	No Moving nonconductive foreigner contamination between two pins	Ignore	Visual	Glue remained on the golden finger	Not Allowed	Visual
Item	Accept Qty	Remark												
Conductive contamination exist in PINs	Not Allowed	Visual												
No Moving nonconductive foreigner contamination between two pins	Ignore	Visual												
Glue remained on the golden finger	Not Allowed	Visual												

4)	Circuit Crack	 Circuit crack---Not Allowed <table border="1"> <thead> <tr> <th>Item</th><th>Accept Qty</th><th>Remark</th></tr> </thead> <tbody> <tr> <td>Circuit crack</td><td>Not Allowed</td><td>Check by CCD or Microscope</td></tr> </tbody> </table>	Item	Accept Qty	Remark	Circuit crack	Not Allowed	Check by CCD or Microscope
Item	Accept Qty	Remark						
Circuit crack	Not Allowed	Check by CCD or Microscope						
5)	Golden finger discoloration	 Golden finger no discoloration ---OK  Golden finger turns black ---NG <table border="1"> <thead> <tr> <th>Item</th><th>Accept Qty</th><th>Remark</th></tr> </thead> <tbody> <tr> <td>Golden finger discoloration</td><td>Copper exposure, turns black (as upper photo) is not allowed.</td><td>Check by 10 time Microscope</td></tr> </tbody> </table>	Item	Accept Qty	Remark	Golden finger discoloration	Copper exposure, turns black (as upper photo) is not allowed.	Check by 10 time Microscope
Item	Accept Qty	Remark						
Golden finger discoloration	Copper exposure, turns black (as upper photo) is not allowed.	Check by 10 time Microscope						
6)	Scratches on FPC coverlayer	 No copper layer exposed---Accepted <table border="1"> <thead> <tr> <th>Item</th><th>Accept Qty</th><th>Remark</th></tr> </thead> <tbody> <tr> <td>Scratches on the coverlayer but no copper layer exposed</td><td>Ignore</td><td>Check by 10 time Microscope</td></tr> </tbody> </table>	Item	Accept Qty	Remark	Scratches on the coverlayer but no copper layer exposed	Ignore	Check by 10 time Microscope
Item	Accept Qty	Remark						
Scratches on the coverlayer but no copper layer exposed	Ignore	Check by 10 time Microscope						
7)	FPC Crease	Photos for dead fold:   Coverlayer turns white--- Not Allowed						

		Photos for not dead fold:   Crease existed, but coverlayer not turns white---Accept <table border="1"> <thead> <tr> <th>Item</th><th>Accept Qty</th><th>Remark</th></tr> </thead> <tbody> <tr> <td>Dead fold</td><td>Not Allowed</td><td>FPC coverlayer will turns white for the dead fold parts, to check by 10 time microscope if need.</td></tr> </tbody> </table>	Item	Accept Qty	Remark	Dead fold	Not Allowed	FPC coverlayer will turns white for the dead fold parts, to check by 10 time microscope if need.									
Item	Accept Qty	Remark															
Dead fold	Not Allowed	FPC coverlayer will turns white for the dead fold parts, to check by 10 time microscope if need.															
8)	EMI shielding Film (From IPC-A-600H 4.1.13)	 图4113a  图4113b  图4113c Acceptable - Class 1,2,3 (可接受条件 - 1.2.3级) • Voids, scratches or foreign material do not exceed 5 locations per side. • Voids, scratches or foreign material do not expose metal underneath the silver film (exposed coverlay underneath is acceptable). • Voids are less than 3.0 x 3.0 mm [0.118 x 0.118 in] in size. • The width of scratches does not exceed 1.5 mm [0.060 in], and their length does not exceed 50% of the length of the flexible printed board. • Touch up by black oil pen (see Figure 4113c). • 每面的空洞、划痕或外来夹杂物不超过5处。 • 空洞、划痕或外来夹杂物未暴露银膜下的金属（暴露银膜层可接受）。 • 空洞尺寸小于3.0x3.0mm[0.118x0.118in]。 • 划痕宽度未超过1.5mm[0.060in]，且长度未超过挠性印制板长度的50%。 • 用黑色油性笔进行修补（见图4113c）。 <table border="1"> <thead> <tr> <th>Item</th><th>Accept Qty</th><th>Remark</th></tr> </thead> <tbody> <tr> <td>Exceed 5 locations per side</td><td>Not Allowed</td><td>Visual</td></tr> <tr> <td>Exposed metal underneath(Cu layer)</td><td>Not Allowed</td><td>Visual</td></tr> <tr> <td>The diameter of voids exceeds 3.0mm.</td><td>Not Allowed</td><td>Visual</td></tr> <tr> <td>The width of scratches exceeds 1.5mm, or length exceeds 50% of the length of FPC.</td><td>Not Allowed</td><td>Visual</td></tr> </tbody> </table>	Item	Accept Qty	Remark	Exceed 5 locations per side	Not Allowed	Visual	Exposed metal underneath(Cu layer)	Not Allowed	Visual	The diameter of voids exceeds 3.0mm.	Not Allowed	Visual	The width of scratches exceeds 1.5mm, or length exceeds 50% of the length of FPC.	Not Allowed	Visual
Item	Accept Qty	Remark															
Exceed 5 locations per side	Not Allowed	Visual															
Exposed metal underneath(Cu layer)	Not Allowed	Visual															
The diameter of voids exceeds 3.0mm.	Not Allowed	Visual															
The width of scratches exceeds 1.5mm, or length exceeds 50% of the length of FPC.	Not Allowed	Visual															
9)	Other Failures	To sign the golden sample if need															

11.9.2. Electrical defect

Item #	Inspection Item	Acceptance Criteria	
01	Dot(Pixel Defect)	Defected item	Active area
		Bright Dot	$N \leq 0$
		Dark Dot	$N \leq 3$
		Total Dot	$N \leq 4$
		Two Adjacent Dot	$N \leq 1$
		Three or More Adjacent Dot	Not Allowed
		Line Defect	Not Allowed
		Mura	Accept if it can not be visible by 5% ND filter in 50% gray pattern. Refer to limit sample if need.
		Remark:	
		a. One pixel consists of 3 sub-pixels, including R,G and B dot(Sub-pixel=Dot)	
b. The defective area of the dot is larger than 50% of one sub-pixel area as one defect; less than 50% of one sub-pixel area will be not made as one failure.			
c. Bright dot is defined through 5% transmission ND filter as following:			
			
d. Judge defect dot and adjacent dot as following. Allow below (as A, B, C and D status) adjacent defect dots, including bright and dark adjacent dot. And they will be counted as 2 defect dots in total quantity.			
			
Defects on the Black Matrix, out of Display area, are not considered as a defect or counted.			
02	Electrical Defect	The below defects must be rejected. 1) Missing vertical / horizontal segment, 2) Abnormal Display. 3) No function or no display. 4) Current exceeds product specifications. 5) LCD viewing angle defect. 6) No Backlight. 7) Dark Backlight.	

12. Identification/marketing criteria:

Any unit with illegible / wrong /double or no marking/ label shall be rejected.

13. Packing:

There should be no damage of the outside carton box, each packaging box should have one identical label.

Modules inside package box should have compliant mark.

All direct package materials shall offer ESD protection.

14. Reliability Specification

Item	Condition	Cycle Time	Quantity	Remark
High Temp. Operation Test	+70 °C	96hrs	5pcs	
Low Temp. Operation Test	-20 °C	96hrs	5pcs	
High Temp. Storage Test	+80 °C	96hrs	5pcs	
Low Temp. Storage Test	-30 °C	96hrs	5pcs	
High Temperature and High Humidity(operation)	Ta=+50 °C, 90%RH	96 hrs	5pcs	
Thermal Shock Test	-30°C (30min) → +80°C (30min)	10cycles	5pcs	
Laminative load test (for packaging)	First place the sample carton on a plane, then load the standard weight object on the top of the sample carton, finally observe the status of the sample carton	24 hrs	One inner carton	Refer to NVD WI Document, WI-Q4 L-0196
Packing Drop test (for packaging)	1 drop on a corner, 1 drop on three arris, 1 drop on six sides	1time	One inner carton	
ESD	150pF,330Ω, ± 4KV & ±6KV contact & air discharge at the center of the screen	10times	5pcs	

Note: 1.For humidity test, DI water should be used.

Inspection Standard: Inspect after 1-2hrs storage at room temperature, the sample shall be free from the following defects:

- Air bubble in the LCD
- Seal Leakage
- Non-display
- Missing Segment
- Glass Crack
- IDD is greater than twice initial value.
- Others as per QA Inspection Criteria

2. No defect is allowed after testing.

3. ESD should be applied to LCD glass panel, not other areas (such as on IC and so on)
IDD should be within twice initial value.

In case of malfunction defect caused by ESD damage, if it would be recovered to normal state after resetting, it would be judged as a good part.

15. Precautions and Warranty

15.1. Safety

- 15.1.1. The liquid crystal in the LCD is poisonous. Do not put it in your mouth. If the liquid crystal touches your skin or clothes, wash it off immediately using soap and water.
- 15.1.2. Since the liquid crystal cells are made of glass, do not apply strong impact on them. Handle with care.

15.2. Handling

- 15.2.1. Reserve and use within ratings in order to keep performance and prevent damage.
- 15.2.2. Do not wipe the polarizer with dry cloth, as it might cause scratch. If the surface of the LCD needs to be cleaned, wipe it swiftly with cotton or other soft cloth soaked with petroleum IPA, do not use other chemicals.

15.3. Storage

- 15.3.1. Do not store the LCD module beyond the specified temperature ranges.

15.4. Metal Pin (Apply to Products with Metal Pins)

15.4.1. Pins of LCD and Backlight

- 15.4.1.1. Solder tip can touch and press on the tip of Pin LEAD during the soldering

15.4.1.2. Recommended Soldering Conditions

Solder Type: Sn96.3~94-Ag3.3~4.3-Cu0.4~1.1

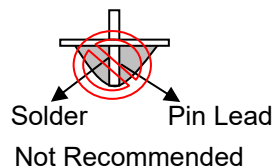
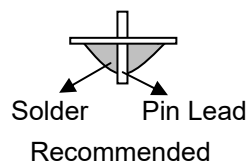
Maximum Solder Temperature: 370°C

Maximum Solder Time: 3s at the maximum temperature

Recommended Soldering Temp: 350±20°C

Typical Soldering Time: ≤3s

15.4.1.3. Solder Wetting



15.4.2. Pins of EL

- 15.4.2.1. Solder tip can touch and press on the tip of EL leads during soldering.

- 15.4.2.2. No Solder Paste on the soldering pad on the motherboard is recommended.

15.4.2.3. Recommended Soldering Conditions

Solder type: Nippon Alimit Leadfree SR-34, size 0.5mm

Recommended Solder Temperature: 270~290°C

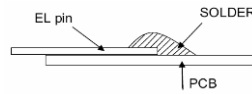
Typical Soldering Time: ≤2s

Minimum solder distance from EL lamp (body): 2.0mm

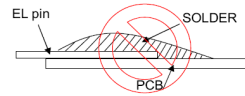
15.4.2.4. No horizontal press on the EL leads during soldering.

15.4.2.5. 180° bend EL leads three times is not allowed.

15.4.2.6. Solder Wetting

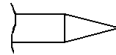


Recommended

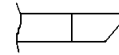


Not Recommended

15.4.2.7. The type of the solder iron:

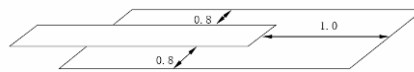


Recommended



Not Recommended

15.4.2.8. Solder Pad



15.5. Operation

15.5.1. Do not drive LCD with DC voltage

15.5.2. Response time will increase below lower temperature

15.5.3. Display may change color with different temperature

15.5.4. Mechanical disturbance during operation, such as pressing on the display area, may cause the segments to appear "fractured".

15.6. Static Electricity

15.6.1. CMOS LSIs are equipped in this unit, so care must be taken to avoid the electro-static charge, by ground human body, etc.

15.6.2. The normal static prevention measures should be observed for work clothes and benches.

15.6.3. The module should be kept into anti-static bags or other containers resistant to static for storage.

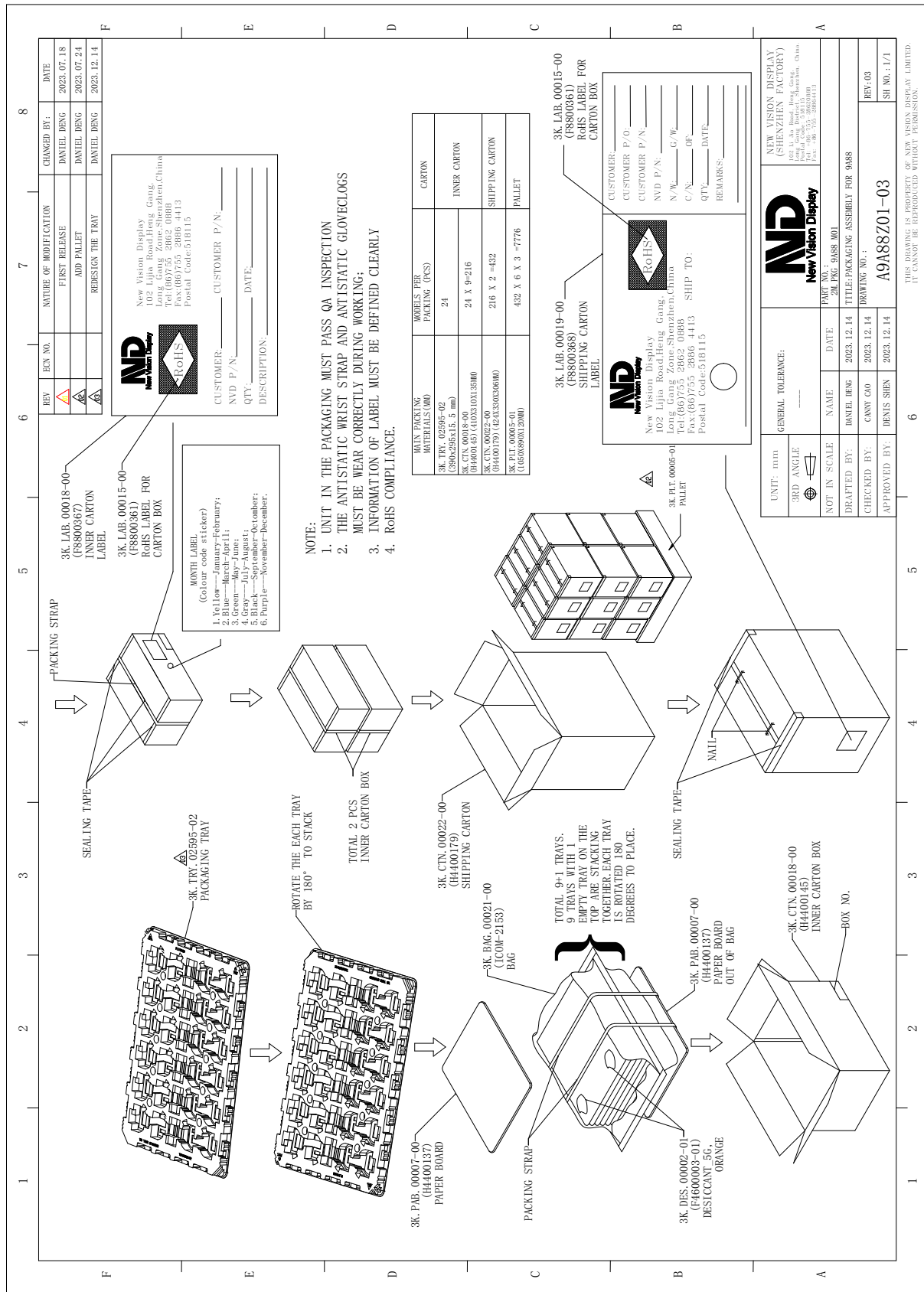
15.7. Limited Warranty

15.7.1. Unless otherwise agreed between New Vision Display and customer, New Vision Display will replace or repair any of its LCD and LCM which New Vision Display found to be defective electrically and visually when inspected in accordance with New Vision Display Quality Standards, for a period of one year from date of shipment.

15.7.2. The warranty liability of New Vision Display is limited to repair and/or replacement. New Vision Display will not be responsible for any consequential loss.

15.7.3. If possible, we suggest you use up all modules in six months. If the module storage time over twelve months, we suggest that recheck it before the module be used.

16. Packaging



1. DISPLAY TYPE: 14.4" FTS/28(RGB)X128 DOTS.
2. HUMAN EYE VIEWING DIRECTION: FULL VIEW.
3. BACKLIGHT: 3 FCS WHITE LED.
4. OVER VOLTAGE: 12V/12.5V/13V/13.5V/14V/14.5V/15V/15.5V/16V/16.5V/17V/17.5V/18V/18.5V/19V/19.5V/20V/20.5V/21V/21.5V/22V/22.5V/23V/23.5V/24V/24.5V/25V/25.5V/26V/26.5V/27V/27.5V/28V/28.5V/29V/29.5V/30V/30.5V/31V/31.5V/32V/32.5V/33V/33.5V/34V/34.5V/35V/35.5V/36V/36.5V/37V/37.5V/38V/38.5V/39V/39.5V/40V/40.5V/41V/41.5V/42V/42.5V/43V/43.5V/44V/44.5V/45V/45.5V/46V/46.5V/47V/47.5V/48V/48.5V/49V/49.5V/50V/50.5V/51V/51.5V/52V/52.5V/53V/53.5V/54V/54.5V/55V/55.5V/56V/56.5V/57V/57.5V/58V/58.5V/59V/59.5V/60V/60.5V/61V/61.5V/62V/62.5V/63V/63.5V/64V/64.5V/65V/65.5V/66V/66.5V/67V/67.5V/68V/68.5V/69V/69.5V/70V/70.5V/71V/71.5V/72V/72.5V/73V/73.5V/74V/74.5V/75V/75.5V/76V/76.5V/77V/77.5V/78V/78.5V/79V/79.5V/80V/80.5V/81V/81.5V/82V/82.5V/83V/83.5V/84V/84.5V/85V/85.5V/86V/86.5V/87V/87.5V/88V/88.5V/89V/89.5V/90V/90.5V/91V/91.5V/92V/92.5V/93V/93.5V/94V/94.5V/95V/95.5V/96V/96.5V/97V/97.5V/98V/98.5V/99V/99.5V/100V/100.5V/101V/101.5V/102V/102.5V/103V/103.5V/104V/104.5V/105V/105.5V/106V/106.5V/107V/107.5V/108V/108.5V/109V/109.5V/110V/110.5V/111V/111.5V/112V/112.5V/113V/113.5V/114V/114.5V/115V/115.5V/116V/116.5V/117V/117.5V/118V/118.5V/119V/119.5V/120V/120.5V/121V/121.5V/122V/122.5V/123V/123.5V/124V/124.5V/125V/125.5V/126V/126.5V/127V/127.5V/128V/128.5V/129V/129.5V/130V/130.5V/131V/131.5V/132V/132.5V/133V/133.5V/134V/134.5V/135V/135.5V/136V/136.5V/137V/137.5V/138V/138.5V/139V/139.5V/140V/140.5V/141V/141.5V/142V/142.5V/143V/143.5V/144V/144.5V/145V/145.5V/146V/146.5V/147V/147.5V/148V/148.5V/149V/149.5V/150V/150.5V/151V/151.5V/152V/152.5V/153V/153.5V/154V/154.5V/155V/155.5V/156V/156.5V/157V/157.5V/158V/158.5V/159V/159.5V/160V/160.5V/161V/161.5V/162V/162.5V/163V/163.5V/164V/164.5V/165V/165.5V/166V/166.5V/167V/167.5V/168V/168.5V/169V/169.5V/170V/170.5V/171V/171.5V/172V/172.5V/173V/173.5V/174V/174.5V/175V/175.5V/176V/176.5V/177V/177.5V/178V/178.5V/179V/179.5V/180V/180.5V/181V/181.5V/182V/182.5V/183V/183.5V/184V/184.5V/185V/185.5V/186V/186.5V/187V/187.5V/188V/188.5V/189V/189.5V/190V/190.5V/191V/191.5V/192V/192.5V/193V/193.5V/194V/194.5V/195V/195.5V/196V/196.5V/197V/197.5V/198V/198.5V/199V/199.5V/200V/200.5V/201V/201.5V/202V/202.5V/203V/203.5V/204V/204.5V/205V/205.5V/206V/206.5V/207V/207.5V/208V/208.5V/209V/209.5V/210V/210.5V/211V/211.5V/212V/212.5V/213V/213.5V/214V/214.5V/215V/215.5V/216V/216.5V/217V/217.5V/218V/218.5V/219V/219.5V/220V/220.5V/221V/221.5V/222V/222.5V/223V/223.5V/224V/224.5V/225V/225.5V/226V/226.5V/227V/227.5V/228V/228.5V/229V/229.5V/230V/230.5V/231V/231.5V/232V/232.5V/233V/233.5V/234V/234.5V/235V/235.5V/236V/236.5V/237V/237.5V/238V/238.5V/239V/239.5V/240V/240.5V/241V/241.5V/242V/242.5V/243V/243.5V/244V/244.5V/245V/245.5V/246V/246.5V/247V/247.5V/248V/248.5V/249V/249.5V/250V/250.5V/251V/251.5V/252V/252.5V/253V/253.5V/254V/254.5V/255V/255.5V/256V/256.5V/257V/257.5V/258V/258.5V/259V/259.5V/260V/260.5V/261V/261.5V/262V/262.5V/263V/263.5V/264V/264.5V/265V/265.5V/266V/266.5V/267V/267.5V/268V/268.5V/269V/269.5V/270V/270.5V/271V/271.5V/272V/272.5V/273V/273.5V/274V/274.5V/275V/275.5V/276V/276.5V/277V/277.5V/278V/278.5V/279V/279.5V/280V/280.5V/281V/281.5V/282V/282.5V/283V/283.5V/284V/284.5V/285V/285.5V/286V/286.5V/287V/287.5V/288V/288.5V/289V/289.5V/290V/290.5V/291V/291.5V/292V/292.5V/293V/293.5V/294V/294.5V/295V/295.5V/296V/296.5V/297V/297.5V/298V/298.5V/299V/299.5V/300V/300.5V/301V/301.5V/302V/302.5V/303V/303.5V/304V/304.5V/305V/305.5V/306V/306.5V/307V/307.5V/308V/308.5V/309V/309.5V/310V/310.5V/311V/311.5V/312V/312.5V/313V/313.5V/314V/314.5V/315V/315.5V/316V/316.5V/317V/317.5V/318V/318.5V/319V/319.5V/320V/320.5V/321V/321.5V/322V/322.5V/323V/323.5V/324V/324.5V/325V/325.5V/326V/326.5V/327V/327.5V/328V/328.5V/329V/329.5V/330V/330.5V/331V/331.5V/332V/332.5V/333V/333.5V/334V/334.5V/335V/335.5V/336V/336.5V/337V/337.5V/338V/338.5V/339V/339.5V/340V/340.5V/341V/341.5V/342V/342.5V/343V/343.5V/344V/344.5V/345V/345.5V/346V/346.5V/347V/347.5V/348V/348.5V/349V/349.5V/350V/350.5V/351V/351.5V/352V/352.5V/353V/353.5V/354V/354.5V/355V/355.5V/3